

Article

Research on Variable Universe Fuzzy Multi-Parameter Self-Tuning PID Control of Bridge Crane

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Abstract: The bridge-type bridge crane is a common lifting equipment used in modern factories and workshops. During the crane's operation, the positioning of the trolley and the swing of the load can significantly impact the bridge crane's safety and reliability. In this paper, we propose a variable universe fuzzy multi-parameter self-tuning PID (VUFMS-PID) control strategy for controlling the trolley's movement. Our control strategy uses scaling factor variation to dynamically adjust the number of fuzzy control rules based on the system error and error rate of change. This approach improves control accuracy and enhances the crane's stability and safety. Simulation results demonstrate that our proposed control strategy outperforms both the fuzzy PID and traditional PID control strategies. Specifically, it reduces the crane trolley's positioning time and the maximum swing angle of the load. Our control strategy exhibits good adaptive ability and robustness, which further improves the stability and safety of the bridge-type bridge crane operation.

Keywords: bridge-type bridge crane; fuzzy control; PID control; positioning; swing angle



Citation: Li, H.; Hui, Y.; Ma, J.; Wang, Q.; Zhou, Y.; Wang, H. Research on Variable Universe Fuzzy Multi-Parameter Self-Tuning PID Control of Bridge Crane. *Appl. Sci.* **2023**, *13*, 4830. <https://doi.org/10.3390/app13084830>

Academic Editor: Jordi Cusido

Received: 22 March 2023

Revised: 7 April 2023

Accepted: 10 April 2023

Published: 12 April 2023



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1. Introduction

With the continuous development of automatic control technology, bridge cranes have gradually evolved from a single handling tool to an important component of automation and flexible production, widely used in industrial places such as factories, workshops, warehouses, and ports. However, during the transportation process of the crane, the acceleration and deceleration of the carts and trolleys, the lifting of the load, and the disturbance caused by wind friction can cause the load to swing back and forth. In the past, operators relied on experience to overcome this, which not only increased the possibility of accidents but also did not help improve the efficiency of crane operation.

When it comes to the positioning and anti-sway control of cranes, they can be categorized as open-loop control and closed-loop control based on the type of control. For instance, input-shaping controllers, which are commonly used in [1,2], fall under open-loop control. These controllers are designed to be simple and have a certain impact on anti-sway and positioning. Moreover, engineering implementation is relatively easy. However, when cranes are operating under external disturbances, or when the initial swing angle of the load is not zero, the anti-sway control effect of the open-loop controller can be severely diminished due to the absence of feedback in the input-shaping control system.

As control technology and sensor technology continue to advance, more scholars are focusing on feedback control. In [3], the anti-swing control of a bridge crane was studied using adaptive sliding mode control (SMC), which included compensation and prediction mechanisms. To ensure the stability and robustness of the control system, suitable Lyapunov functions were developed. In a study by Ouyang et al. [4], they proposed an adaptive nonlinear controller using the dynamic model of a double-pendulum rotary crane. By

employing Lyapunov functions, they were able to prove the stability of the entire control system, resulting in successful control outcomes. In [5], an optimal controller was used to control a variable-parameter bridge crane, and Monte Carlo simulation was employed to determine the parameters of the optimal controller. A control approach employing linear active disturbance rejection control (LADRC) and differential flatness theory was proposed in a recent study by Chai et al. [6]. The control method was optimized using particle swarm optimization (PSO) to fine-tune the controller's parameters, resulting in successful positioning and anti-swing control performance. In a study [7], a passivity-based adaptive control approach was introduced for a bridge crane. The authors proved the stability of the closed-loop control system using passivity theory and developed an adaptive control law. Simulation results confirmed the effectiveness of the proposed approach. In [8], a new nonlinear control method and disturbance observer were proposed to study the adjustment and suppression of uncertain disturbances in bridge crane systems. In [9], a model predictive control system was proposed to achieve swing attenuation of the trolley hoist. When encountering external disturbances, the swing angle and load displacement were used as feedback inputs for anti-swing control in the closed-loop system. In [10], a robust linear quadratic regulator (LQR) based on particle swarm optimization (PSO) was used for swing prevention and positioning control of the system. The controller uses a parallel compensator structure and optimizes the parameter selection of the weight matrix using the particle swarm optimization algorithm.

These advanced control methods have demonstrated their effectiveness in achieving accurate positioning and anti-sway control in cranes. However, some controllers have their own drawbacks. For example, the sliding mode controller heavily relies on mathematical models and expert experience in designing the sliding surface and determining its parameters can be difficult. The LQR controller heavily relies on the linear model of the controlled object, and if the system model is inaccurate, it may lead to a decrease in the reliability of the controller. The MPC controller needs to solve an optimization problem in each control cycle, which has a high computational complexity and requires the selection and adjustment of multiple parameters, making it a complicated design process. This makes it difficult for some controllers to be widely applied in practical engineering at present. In contrast, the PID feedback control system, with its reliable and stable performance, has been widely used in crane applications [11–13]. The selection of the three control parameters in the PID system significantly influences the controller's performance, leading many researchers to investigate the optimization of these parameters. For instance, Solihin et al. [12] utilized a genetic algorithm (GA) to optimize the gain of PID, resulting in excellent control performance in both trolley positioning and load anti-sway. Jaafar et al. [14] employed an improved particle swarm optimization (PSO) algorithm, selecting an adaptive fitness function based on the priority fitness scheme (PFS). The PID controller has been shown to effectively control load swings and ensure system stability. Furthermore, in a study by Yan et al. [15], particle swarm optimization (PSO) was employed to optimize the vertical position and speed trajectory of the path planning control system for a crane. The proposed method was tested under different sea conditions using simulations, which demonstrated its effectiveness. Xia et al. [16] employed a dual-closed-loop composite anti-sway control method to regulate both the trolley's position and the hanging load's swing of an overhead crane, optimizing the parameters of the PD controller through the sequential quadratic optimization method. Some researchers have also integrated PID controllers with other control techniques. Yan et al. [17] presented a novel anti-sway control approach that integrated PID control and neural network compensation. They established the stability of the neural PID control system using a standard weight training algorithm and report satisfactory simulation outcomes.

Bridge cranes are non-linear control systems that require accurate mathematical models for precise control when using linear controllers. Fuzzy controllers are based on expert experience without requiring accurate mathematical models, making them very suitable for controlling non-linear systems [18–21]. Combining fuzzy controllers with other controllers

can fully leverage the advantages of both types of controllers, achieving good results in the field of crane control [22–25]. A fuzzy PID controller modifies the PID parameters in real time using fuzzy control rules. It determines the controller's output by considering the magnitude and rate of change of the input variables. This controller is widely used in the field of cranes. In [26], control research was conducted on a bridge crane using a multi-input single-output fuzzy algorithm model combined with PID. Simulation results demonstrated that the fuzzy PID control system exhibits superior robustness compared to the classical PID control system when handling external disturbances. In [27], an approach combined fuzzy and PID controllers and derived heuristic control rules using membership functions. The parameters of the membership functions were then tuned through an adaptive neural fuzzy inference system. This control system achieved faster speed and less swing over longer transport distances. In [28], a combination of PID and fuzzy controllers was used to compensate for the coupling between friction, gravity, and the position of the cart, as well as anti-swing control. In order to enhance the performance of the PD controller, a high-gain observer was incorporated to estimate the cart's speed. Solihin et al. [29] proposed a fuzzy-tuned PID controller design methodology for anti-sway control of bridge cranes, which employed a fuzzy system to adjust the PID controller's parameters. This approach exhibited strong robustness against variations in crane parameters. Simulation outcomes demonstrated that the controller yielded favorable anti-sway performance. However, in the later stages of the operation of the fuzzy PID control system for crane anti-sway control, the feedback input quantity is often small while the input and output domains are large. Therefore, more control rules are required to satisfy the control accuracy, which can be difficult to implement in deterministic fuzzy controllers.

To tackle the challenge of having an increasing number of fuzzy control rules in a fuzzy PID controller, we introduce a variable-domain fuzzy PID controller as an alternative. This controller has the ability to adjust the size of the input and output domains as the input changes, thereby enhancing control accuracy. Simulation comparisons reveal that the proposed variable-domain fuzzy PID controller to achieve superior control performance in terms of crane positioning and anti-sway angle compared to both the traditional PID controller and the fuzzy PID controller. Furthermore, it exhibit robustness in the face of various operating conditions and environmental noise interference.

The remainder of this paper will be structured as follows: Section 2 presents a dynamic model of the bridge crane. In Section 3, we design a variable universe fuzzy controller for the bridge crane. Section 4 conducts simulation and analysis, and Section 5 performs a robustness analysis of the proposed controller. Finally, the concluding chapter summarizes the findings of this study.

2. Bridge Crane Mechanical Dynamics Modeling

2.1. State Assumption and System Simplification

The bridge crane system is highly complex and exhibits strong nonlinearity. To simplify the theoretical analysis, some secondary factors such as wire rope deformation, wire rope mass, and wind resistance are ignored. The three-dimensional model of the crane operation system is then reduced to a two-dimensional single pendulum model of the trolley-load system [23], as shown in Figure 1.

The load is connected to the crane trolley via a rope. When an external force F is applied, the crane trolley moves horizontally and is subject to the frictional force f of the track. The crane trolley has a mass of m_1 , the load has a mass of m_2 , the rope has a length of l , the pendulum angle is θ , and the horizontal displacement is x .

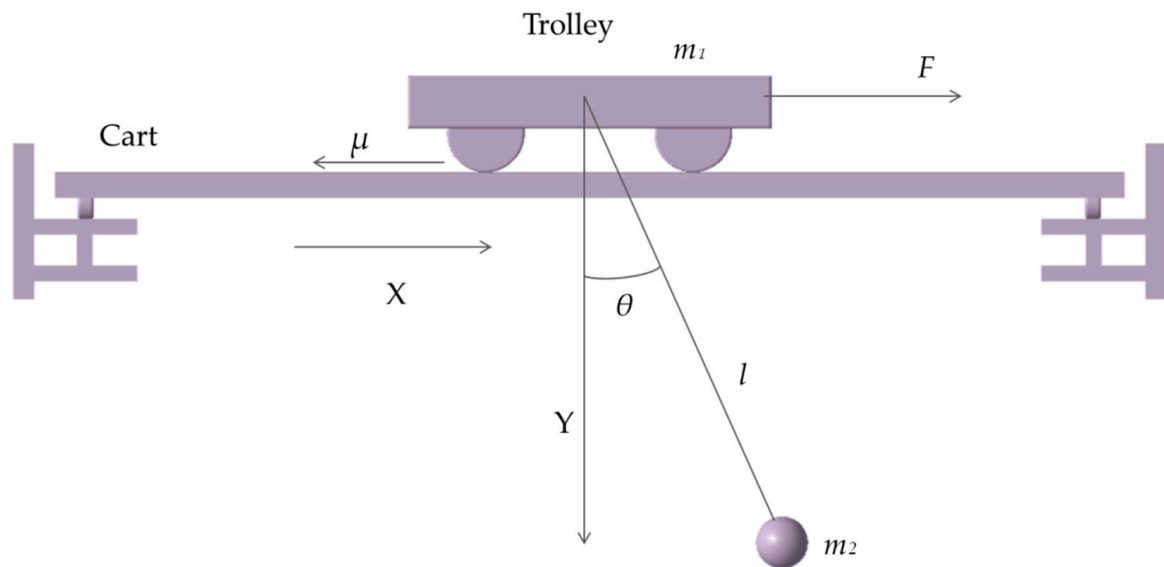


Figure 1. Motion diagram of bridge crane.

2.2. System Dynamics Modeling

Assuming the coordinate of the crane's center of mass is $(x, 0)$, according to the mechanical relationships in Figure 1, the displacement of the bridge crane anti-sway system's trolley and load can be obtained as follows.

$$\begin{cases} x_{m1} = x \\ y_{m1} = 0 \\ x_{m2} = x + l \sin \theta \\ y_{m2} = l \cos \theta \end{cases} \quad (1)$$

In Equation (1), x_{m1} , and y_{m1} are the displacement components of the bridge crane trolley in the X and Y directions, respectively, while x_{m2} and y_{m2} are the displacement components of the load in the X and Y directions. Taking the derivative of Equation (1), the velocity components of the crane trolley and load in the X and Y directions can be obtained as follows [12]:

$$\begin{cases} \dot{x}_{m1} = \dot{x} \\ \dot{y}_{m1} = 0 \\ \dot{x}_{m2} = \dot{x} + \dot{l} \sin \theta + l \dot{\theta} \cos \theta \\ \dot{y}_{m2} = \dot{l} \cos \theta - l \dot{\theta} \sin \theta \end{cases} \quad (2)$$

where $\dot{x}_{m1}$ and $\dot{y}_{m1}$ refer to the speed components of the trolley in the X and Y directions, respectively; $\dot{x}_{m2}$ and $\dot{y}_{m2}$ are the velocity components of the load in the X and Y directions. The kinetic energy T of the bridge crane system is the following:

$$T = \frac{1}{2} m_1 v_{m1}^2 + \frac{1}{2} m_2 v_{m2}^2 \quad (3)$$

where m_1 and m_2 are the masses of the crane and the load respectively, and v_{m1} and v_{m2} are the velocities of the crane and the load. Choosing the reference point for zero potential energy at the position of the crane, the potential energy V of the bridge crane anti-sway system is the following:

$$V = -m_2 g l \cos \theta \quad (4)$$

where g is the acceleration due to gravity, $g = 9.8 \text{ m/s}^2$. Therefore, the Lagrange equation for the operation system of the bridge crane is the following:

$$\begin{cases} L = T - V \\ \frac{d}{dx} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = f_i \quad i = 1, 2 \end{cases} \quad (5)$$

where L is the Lagrangian, q_i is the generalized coordinates with q_1 and q_2 being the generalized coordinates for displacement and angle, respectively, and f_i represents the generalized external forces on the i -th coordinate. The frictional force between the trolley wheels and the track is denoted as f and μ is the friction coefficient. By combining Equations (1) to (5), the nonlinear dynamic differential equations for the bridge crane operating system can be obtained as follows:

$$\begin{cases} (m_1 + m_2)\ddot{x} + m_2\dot{l}\sin\theta + 2m_2l\dot{\theta}\cos\theta + \\ m_2l\ddot{\theta}\cos\theta - m_2\dot{l}^2\sin\theta + \mu\dot{x} = F \\ 2\dot{l}\dot{\theta} + l\ddot{\theta} + \ddot{x}\cos\theta + g\sin\theta = 0 \end{cases} \quad (6)$$

Considering the actual situation, in the process of the bridge crane's trolley movement, the swing angle of the load should be as small as possible, generally not exceeding 10° , so $\sin\theta = \theta$ and $\cos\theta = 1$, at the same time, due to the small swing angle of the load, the length of the rope can be approximately regarded as constant, so $\dot{l} = \ddot{l} = 0$. Substituting the approximate values into Equation (6), the differential equation of the bridge crane operating system can be simplified as the following.

$$\begin{cases} (m_1 + m_2)\ddot{x} + m_2l\ddot{\theta} + \mu\dot{x} = F \\ l\ddot{\theta} + \ddot{x} + g\theta = 0 \end{cases} \quad (7)$$

Taking the Laplace transform of the above equation yields the transfer functions for the positioning of the trolley and the swing angle of the load.

$$\begin{cases} \frac{X(s)}{F(s)} = \frac{ls^2 + g}{m_1ls^4 + \mu ls^3 + (m_1 + m_2)gs^2 + \mu gs} \\ \frac{\theta(s)}{F(s)} = \frac{s^2}{m_1ls^4 + \mu ls^3 + (m_1 + m_2)gs^2 + \mu gs} \end{cases} \quad (8)$$

3. Variable Domain Fuzzy Multi-Parameter Self-Tuning PID Controller

3.1. Variable Domain Fuzzy Control

After determining the structure of a conventional fuzzy controller, its fuzzy control rules, input/output variable fuzzy domains, and fuzzy subset partitions are fixed. However, in situations where the input error and the error change rate are both small around the steady state, the fuzzy subset partition of the input variable fuzzy domain may be too coarse compared to the small error. This can result in the controller repeatedly using only a few control rules, which can negatively impact control precision at the steady state.

To address this issue and achieve the desired output without altering the fuzzy subset partition or fuzzy control rules, the input/output variable fuzzy domains must be adjusted in accordance with changes in the error. Specifically, when the error is large, the basic fuzzy domain is appropriately expanded, while it is compressed when the error is small. This approach is equivalent to adding fuzzy control rules and ensuring output accuracy.

Assuming the initial fuzzy domains for the input error e and error change rate ec are $[-E, E]$ and $[-EC, EC]$, respectively, and the initial fuzzy domain for the output variable u is $[-U, U]$. By introducing domain scaling factors, the domains can be represented as:

$$X(e) = [-\alpha(e)E, \alpha(e)E] \quad (9)$$

$$X(ec) = [-\alpha(ec)E, \alpha(ec)E] \quad (10)$$

$$Y(y) = [-\beta(y)U, \beta(y)U] \quad (11)$$

The variables $\alpha(e)$, $\alpha(ec)$, and $\beta(y)$ represent the domain scaling factors for the input variables e and ec , and the output variable u , respectively.

3.2. Bridge Crane Control Scheme

During crane movement, the trolley's displacement and velocity, swing angle, and angular velocity constantly fluctuate. The fuzzy controller can adjust the PID control parameters by monitoring these parameters' changes to improve control performance. To achieve precise control during anti-sway operations, a PD controller is employed due to its ability to eliminate steady-state errors. On the other hand, trolley positioning control utilizes a PID controller for improved accuracy. The displacement s and swing angle θ are used as inputs for the two controllers, respectively. The controllers' outputs are linearly combined to produce the overall controller output, as illustrated in Figure 2.

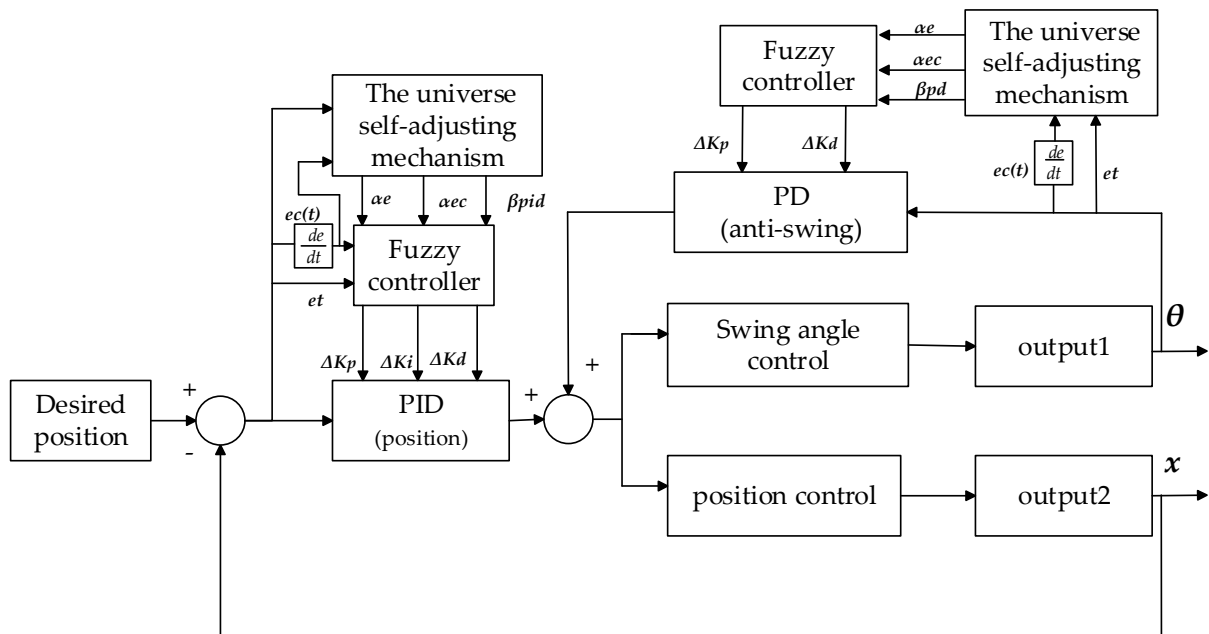


Figure 2. The control scheme for the bridge crane.

The process of domain adjustment involves calculating scaling factors $\alpha(e)$ and $\alpha(ec)$ for the input variables e and ec , respectively, as well as scaling factor $\beta_{(PID)}$ for the output variables ΔK_P , ΔK_I and ΔK_D . These factors are determined based on the deviation e and deviation change rate ec to adjust the size of the input and output domains. Once the scaling factors are determined, the fuzzy controllers perform fuzzification, fuzzy inference, and defuzzification to obtain the output variables ΔK_P , ΔK_I and ΔK_D . The resulting output parameter values are then added to the initial values of K_P , K_I and K_D for the PID controller. This process leads to achieving precise control of the controlled object using the self-tuned K_P , K_I and K_D values.

3.3. Determining the Domain Scaling Factors

In this article, the method of constructing domain scaling factors based on functions is utilized. The selected function must satisfy five conditions: coordination, monotonicity, regularity, duality, and avoidance of zero. Based on the above analysis, the function defined as satisfying these five conditions for domain scaling factors is the following.

$$\begin{cases} \alpha(x) = 1 - \gamma \exp(-kx^2) \\ \beta(y) = K \sum_{i=1}^n p_i \int_0^t e_i(\tau) d\tau + \beta(0) \end{cases} \quad (12)$$

In Equation (12) [30], x is the input variable; γ is the function coefficient, usually $\gamma \in (0,1)$, as γ becomes larger, $\alpha(x)$ becomes smaller, but the change in $\alpha(x)$ is more dramatic, which can lead to the compression of the universe and the rapid response of the system; k is the exponent coefficient and $k > 0$, the larger k , the larger $\alpha(x)$. K is the coefficient of the integral term; n is the number of input variables, taken as 2 here; p_i is the i -th element in the constant vector; e_i is the i -th element in the input deviation vector; $\beta(0)$ is the initial value of the output variable domain scaling factor.

Drawing on the aforementioned principles and the results of multiple experiments, the following formulas have been derived to determine the input domain scaling factors for the positioning and anti-sway fuzzy controllers:

$$\alpha(e) = 1 - 0.65 \exp(-0.5e^2) \quad (13)$$

$$\alpha(ec) = 1 - 0.6 \exp(-0.5ec^2) \quad (14)$$

After analyzing the control effects of K_P , K_I and K_D on the system, the domain scaling factors for the output variables have been determined. It has been found that the monotonicity of K_P and K_D is consistent with that of $|e|$. Conversely, the monotonicity of K_I is opposite to that of $|e|$. After conducting multiple experiments, the following scaling factors have been selected:

$$\beta_{1P} = 0.4|e|, \beta_{1D} = 6|e|, \beta_{1I} = 1/(|e| + 0.5) \quad (15)$$

$$\beta_{2P} = 15|e|, \beta_{2D} = 0.5|e| \quad (16)$$

Specifically, β_{1P} , β_{1D} and β_{1I} denote the scaling factors for the output variable domain of the position PID controller, while β_{2P} and β_{2D} correspond to the scaling factors for the output variable domain of the anti-swing PD fuzzy controller.

3.4. Fuzzy Control Rules Determination

The foundation for implementing fuzzy control is the fuzzy control rule table. In practice, the fuzzy control rules are usually designed based on expert experience. Considering the importance of parameters like proportional, integral, and derivative in PID control, the following adjustment principles have been formulated:

- a. For large differences $|e|$ between the desired and actual values, a larger K_P parameter is selected to quickly reach the predetermined position. To avoid issues such as sudden changes in the rate of change and integral saturation, smaller K_I and K_D parameters are used.
- b. When the difference $|e|$ and the rate of change of the difference $|ec|$ are moderate, smaller K_P and K_I parameters are selected to prevent excessive overshoot and other problems. At the same time, a moderate K_D parameter is selected to balance the response speed of the crane. By doing so, the crane's motion can be controlled smoothly and accurately.
- c. When the error $|e|$ is small, the focus is on maintaining a stable operating state, so smaller K_P and K_I parameters are selected. However, in order to prevent load oscillation and improve the crane's anti-interference ability, a larger K_D parameter is selected when the rate of change of the error $|ec|$ is small. This is because a larger K_D can help to dampen any oscillations that may occur. Conversely, when the rate of change of the error $|ec|$ is large, a smaller K_D parameter is selected to prevent overshoot and instability.

Based on the above three parameter adjustment rules, the input and output variables of the fuzzy PID controller are partitioned into 7 levels. Each represents a fuzzy subset: negative big (NB), negative medium (NM), negative small (NS), zero (Z), positive small

(PS), positive medium (PM), and positive big (PB). These fuzzy subsets are utilized to denote the linguistic values of the input and output variables, and the fuzzy control rules are formulated based on them.

The ΔK_P , ΔK_I and ΔK_D values obtained through fuzzy rules reasoning are represented in the following Tables 1–3 where the fuzzy control rules for ΔK_P , ΔK_I and ΔK_D are shown [31].

Table 1. Fuzzy control rules for ΔK_P .

ΔK_P	ec							
		NB	NM	NS	Z	PS	PM	PB
e	NB	PB	PB	PM	PM	PS	PS	Z
	NM	PB	PB	PM	PM	PS	Z	Z
	NS	PM	PM	PM	PS	Z	NS	NM
	Z	PM	PS	PS	Z	NS	NM	NM
	PS	PM	PS	Z	NS	NS	NM	NM
	PM	Z	Z	NS	NM	NM	NM	NB
	PB	Z	NS	NS	NM	NM	NB	NB

Table 2. Fuzzy control rules for ΔK_I .

ΔK_I	ec							
		NB	NM	NS	Z	PS	PM	PB
e	NB	NB	NB	NB	NM	NM	Z	Z
	NM	NB	NB	NM	NM	NS	Z	Z
	NS	NM	NM	NS	NS	Z	PS	PS
	Z	NM	NS	NS	Z	PS	PS	PM
	PS	NMS	NS	Z	PS	PS	PM	PM
	PM	NS	NS	Z	PS	PS	PM	PB
	PB	Z	Z	PS	PM	PM	PB	PB

Table 3. Fuzzy control rules for ΔK_D .

ΔK_D	ec							
		NB	NM	NS	Z	PS	PM	PB
e	NB	NS	NS	Z	Z	Z	PM	NB
	NM	PS	PS	PS	PS	Z	PS	NM
	NS	PB	PB	PM	PS	Z	PS	NM
	Z	PB	PM	PM	PS	Z	PS	NM
	PS	PB	PM	PS	PS	Z	PS	NS
	PM	PM	PS	PS	PS	Z	PS	NS
	PB	NS	Z	Z	Z	PM	PM	Z

From Figures 3–5, we can more intuitively see the quantitative relationship between input e , ec and output ΔK_P , ΔK_I and ΔK_D . To tune the PID and PD controllers of this system, we use the same fuzzy control rules to adjust the gain tuning quantities ΔK_P , ΔK_I and ΔK_D . Triangular membership functions are used for the fuzzy input and output membership functions, and the centroid method is used to defuzzify the fuzzy quantities, which provides us with the PID parameter tuning quantities ΔK_P , ΔK_I and ΔK_D . The initial

values of the PID parameters are K_{P0} , K_{I0} and K_{D0} . The formula for online tuning of the PID parameters is as follows.

$$\begin{cases} K_P = K_{P0} + \Delta K_P \\ K_I = K_{I0} + \Delta K_I \\ K_D = K_{D0} + \Delta K_D \end{cases} \quad (17)$$

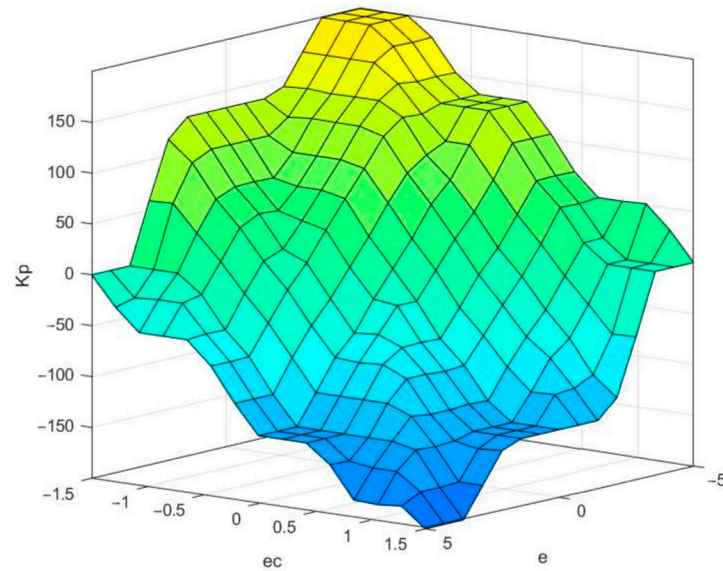


Figure 3. Fuzzy tuner surface of ΔK_P .

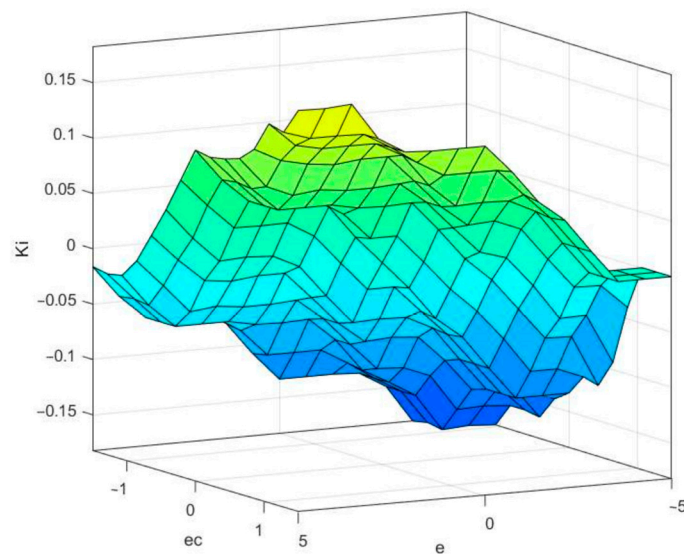


Figure 4. Fuzzy tuner surface of ΔK_I .

The equation above indicates that K_P , K_I and K_D are the PID controller's parameters that are adjusted in real time using the fuzzy control rules for the crane system. On the other hand, K_{P0} , K_{I0} and K_{D0} represent the initial values of the PID controller's parameters for the crane system.

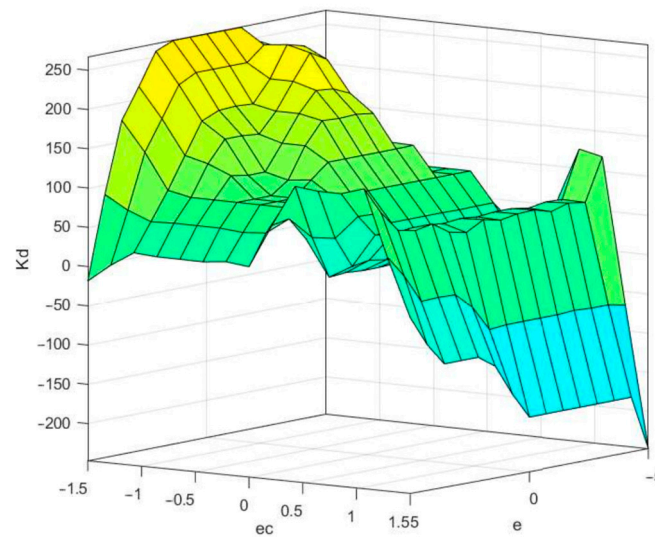


Figure 5. Fuzzy turner surface of ΔK_D .

4. Simulation Analysis

This section presents an evaluation of the performance of three control methods for anti-sway and positioning control of a bridge crane. The methods being evaluated are the traditional PID controller, the FUZZY-PID controller, and the VUFMS-PID controller. To conduct this comparison, we establish a simulation model of the bridge crane using the Simulink module in Matlab. We then subject the three control methods to the same operating conditions and compared their control effects.

In the Matlab simulation, we use a step signal as the control signal and the initial position of the trolley is $X_o = 0$ m, and the target position is $X_d = 0.6$ m. We set the simulation time T to be 30 s and the sampling time T_s to be 0.01 s. The length of the wire rope is $L = 1$ m, the mass of the trolley is $m_1 = 10$ kg, the load mass is $m_2 = 5$ kg, and the gravitational acceleration is $g = 9.8$ m/s² [10]. The basic universe for the controller parameters e and ec is $[-5, 5]$ and $[-1.5, 1.5]$, respectively. The output universe of the parameters ΔK_P , ΔK_I and ΔK_D are $[-200, 200]$, $[-0.2, 0.2]$ and $[-300, 300]$, respectively. Both input and output functions use trigonometric functions. The gains of the positioning PID controller are $K_{1P0} = 16$, $K_{1I0} = 0.1$ and $K_{1D0} = 40$, while the gains of the anti-sway PD controller are $K_{2P0} = 320$ and $K_{2D0} = 100$.

Simulation and Analysis

To compare the effectiveness of the three control methods for the bridge crane's load anti-sway and positioning under the same operating conditions, we evaluated two performance metrics for each method. For positioning control, we evaluated the maximum overshoot (M_p) and settling time of the trolley's position (T_s). For anti-sway control, we evaluated the maximum swing angle (θ_{max}) and stabilization time of the load (T_s). Figures 6 and 7 display the control results of the three methods for the positioning of the trolley and the swing angle of the load, respectively. Table 4 summarizes the performance of each evaluation index for the three control methods.

Based on Figures 6 and 7, it is observed that the VUFMS-PID controller for the bridge crane exhibits the lowest maximum swing angle among the three control methods. Specifically, the maximum swing angle of the load is 0.66 deg for the VUFMS-PID controller, while the FUZZY-PID and PID controllers have maximum swing angles of 0.88 deg and 1.08 deg, respectively. This indicates that the VUFMS-PID controller reduces the maximum swing angle by 0.22 deg and 0.42 deg compared to the FUZZY-PID and PID controllers, respectively, under the same operating conditions. Furthermore, the maximum swing angle of 1.005 deg reported in the literature [10] for a PSO-optimized LQR controller is 0.345 deg (29.1%) higher than that of the VUFMS-PID controller proposed in this study.

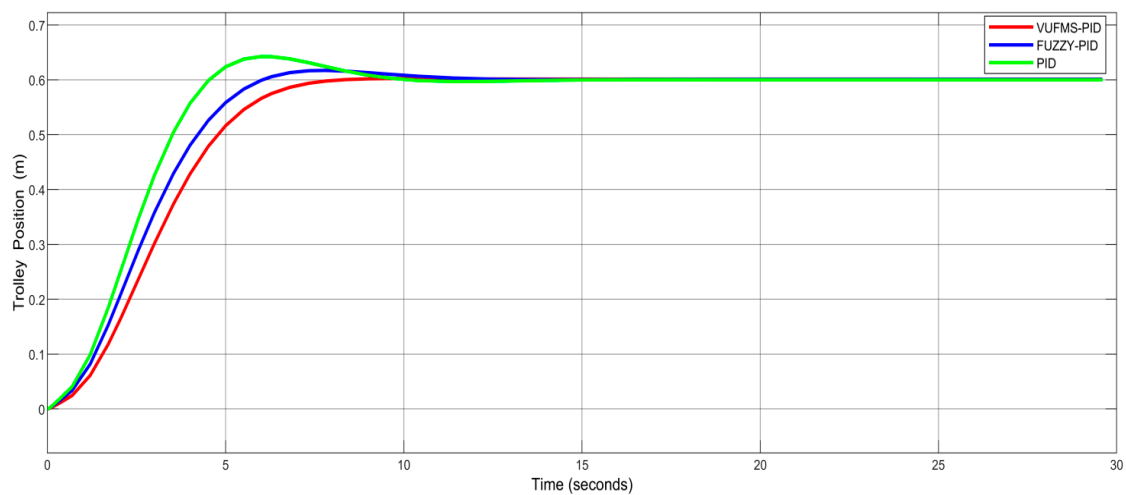


Figure 6. Trolley's position (x) using VUFMS-PID, FUZZY-PID and PID controller.

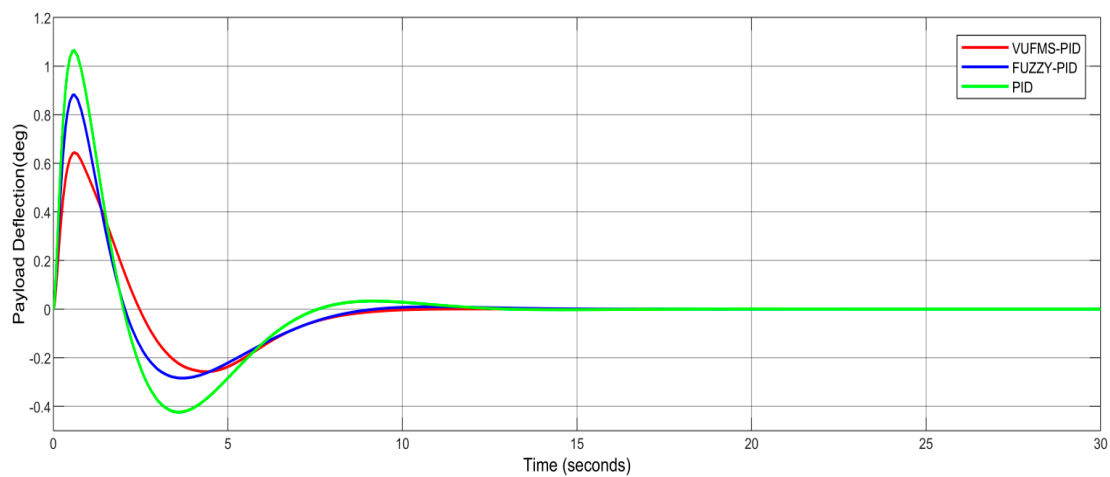


Figure 7. Overhead crane's swing angle (θ) VUFMS-PID, FUZZY-PID and PID controller.

Table 4. Performances obtained using VUFMS-PID, FUZZY-PID and PID controller for the bridge crane.

Output	Performances	Methods			Reference [10]
		VUFMS-PID	FUZZY-PID	PID	
x	T_S (s)	7.9	11.15	11.01	9.3
	M_p (%)	0	2.8	6.8	0
θ	T_S (s)	8.81	9.12	11.85	/
	θ_{max} (deg)	0.66	0.88	1.08	1.005

Regarding the stabilization time of the swing angle, the VUFMS-PID controller has the shortest stabilization time of 8.81 s compared to the FUZZY-PID and PID controllers, which have stabilization times of 9.12 s and 11.85 s, respectively. It is worth noting that literature [10] reported a stabilization time of 9.3 s for swing angle control which is larger than the VUFMS-PID controller's swing angle.

For trolley positioning, the VUFMS-PID controller exhibits the shortest stabilization time of 7.9 s, followed by the FUZZY-PID and PID controllers with stabilization times of 11.15 s and 11.01 s, respectively. In addition, the VUFMS-PID controller has zero overshoot for trolley positioning, while the FUZZY-PID and PID controllers exhibit varying degrees of overshoot.

Based on the above analysis, it can be concluded that the bridge crane system controlled by the VUFMS-PID controller has the smallest overshoot, the smallest swing angle, and the shortest stabilization time for trolley positioning compared to the FUZZY-PID controller, PID controller, and LQR controller proposed in the literature [10], under the same operating conditions. As a result of the evaluation, it can be concluded that the VUFMS-PID controller exhibits the best control performance for the bridge crane system.

5. Robustness Analysis

5.1. Experimental and Simulation Analysis under No Disturbance Conditions

In the previous section, the performance analysis of the crane controller is based on a reference working condition, where the crane lifts a 5 kg load with a 1m rope length. However, in actual working conditions, the load and rope length of the crane are subject to constant variation. To validate the control effectiveness of the proposed VUFMS-PID controller, it is essential to test it under various working conditions of the crane. The following Table 5 presents three additional working conditions with varying loads and rope lengths, which were used to evaluate the control effect of the three controllers.

Table 5. Parameters of the crane under the other three working conditions.

Working Condition	Rope Length (m)	Payload Weight (kg)	Trolley Weight (kg)
1	1.5	5	10
2	1.5	8	10
3	1	8	10

From Figures 8–13 and Tables 6–8, it can be observed that the VUFMS-PID controller for the lifting system exhibits the smallest overshoots of 1%, 0.1% and 0.6% for trolley positioning. The trolley positioning time is also the smallest when compared to the results obtained by the other two controllers. Moreover, the VUFMS-PID controller for the lifting system has the smallest overshoots of 0.58° , 0.56° and 0.57° for the trolley lifting angle. These values are the smallest compared to the other two controllers, and the lifting stability time is also the smallest at 7.86 s, 8.89 s and 10.13 s, respectively. Based on the above analysis, it can be concluded that the proposed VUFMS-PID controller has better adaptability than the FUZZY-PID controller and the traditional PID controller. It can adapt well to the multi-work conditions of the lifting system and achieve good control effects.

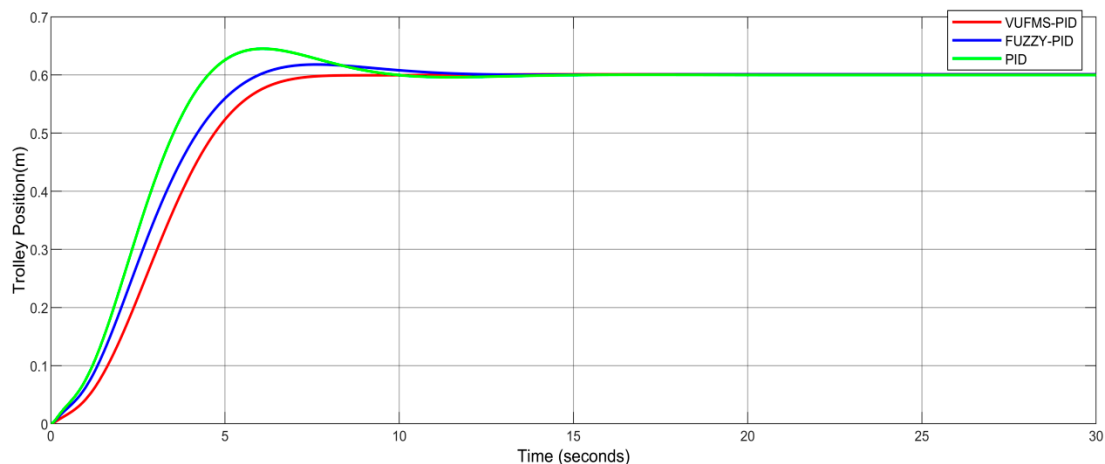


Figure 8. Trolley's position using VUFMS-PID, FUZZY-PID and PID controller under working condition 1.

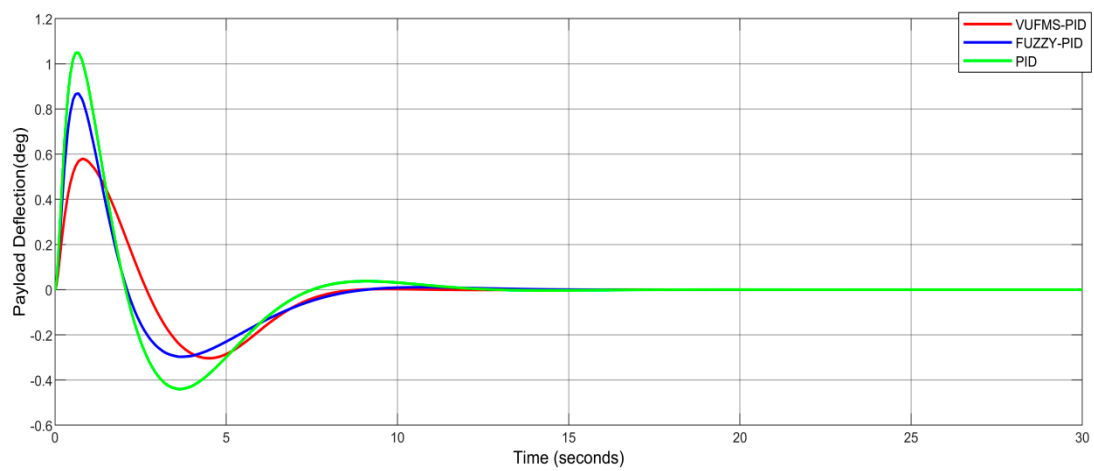


Figure 9. Payload's swing angle (θ) using VUFMS-PID, FUZZY-PID and PID controller under working condition 1.

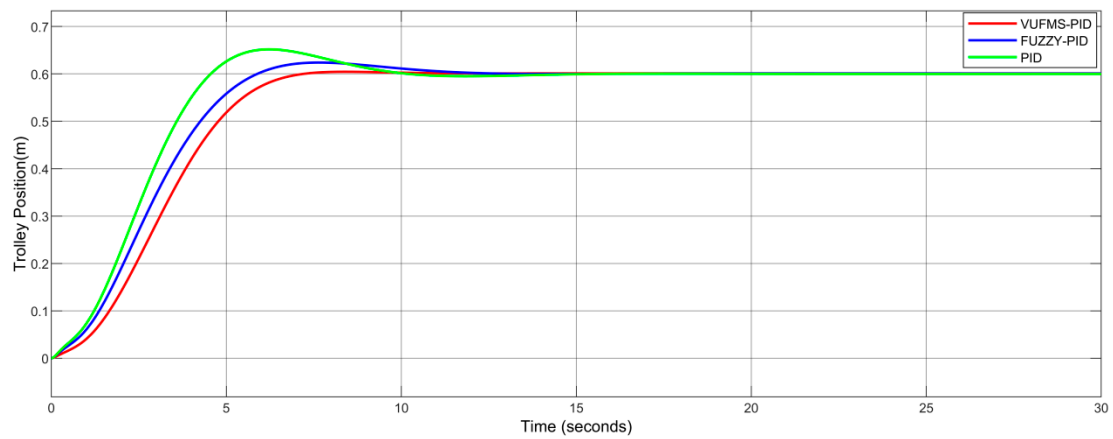


Figure 10. Trolley's position using VUFMS-PID, FUZZY-PID and PID controller under working condition 2.

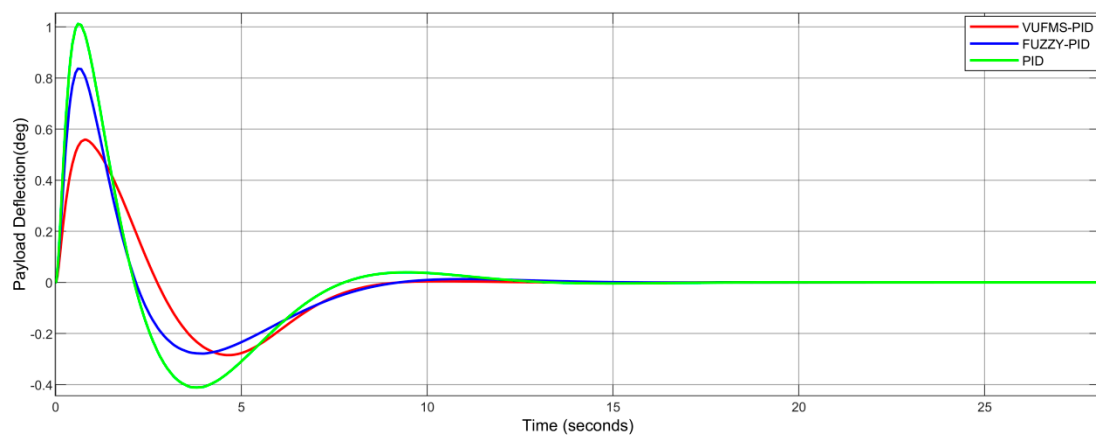


Figure 11. Payload's swing angle (θ) using VUFMS-PID, FUZZY-PID and PID controller under working condition 2.

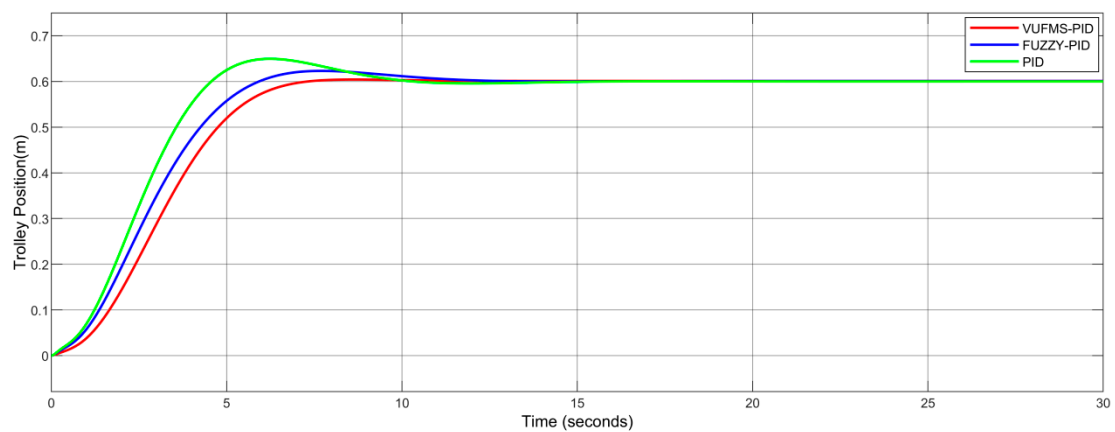


Figure 12. Trolley's position using VUFMS-PID, FUZZY-PID and PID controller under working condition 3.

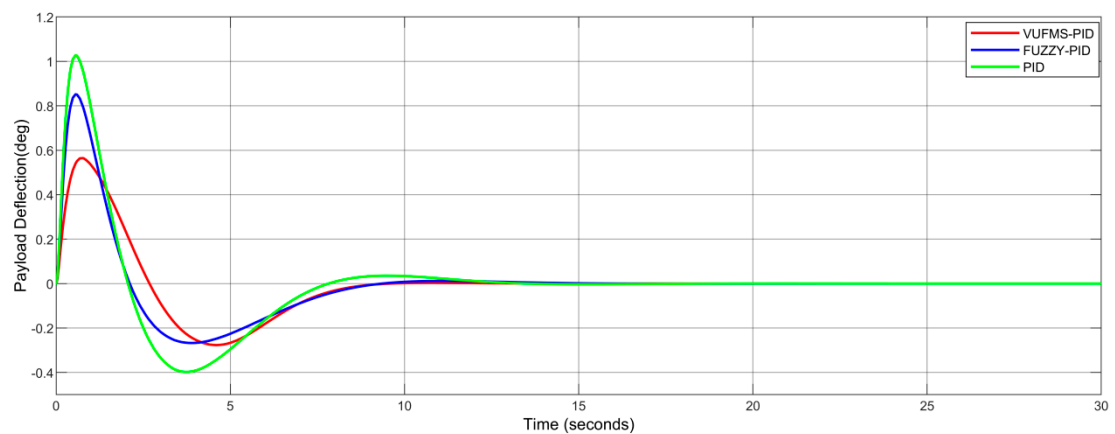


Figure 13. Payload's swing angle (θ) using VUFMS-PID, FUZZY-PID and PID controller under working condition 3.

Table 6. Performances obtained using VUFMS-PID, FUZZY-PID, and PID controller under working condition 1.

Output	Performances	Methods		
		VUFMS-PID	FUZZY-PID	PID
x	T_S (s)	7.86	12.32	10.47
	M_P (%)	0	3.0	6.2
θ	T_S (s)	8.61	11.1	12.05
	θ_{max} (°)	0.58	0.87	1.06

Table 7. Performances obtained using VUFMS-PID, FUZZY-PID, and PID controller under working condition 2.

Output	Performances	Methods		
		VUFMS-PID	FUZZY-PID	PID
x	T_S (s)	8.89	12.13	10.05
	M_P (%)	0.1	4.0	7.3
θ	T_S (s)	9.11	10.81	12.61
	θ_{max} (°)	0.56	0.87	1.01

Table 8. Performances obtained using VUFMS-PID, FUZZY-PID, and PID controller under working condition 3.

Output	Performances	Methods		
		VUFMS-PID	FUZZY-PID	PID
x	T_S (s)	10.13	11.88	11.05
	M_P (%)	0.6	3.8	7.0
θ	T_S (s)	9.38	10.56	12.05
	θ_{max} (°)	0.57	0.84	1.03

5.2. Experimental Simulation Analysis under Disturbance Conditions

To verify the robustness of the controller in controlling the crane under interference, a simulation is conducted. The crane may encounter sudden interferences such as strong winds or environmental noise when working outdoors, hence it is crucial to test the robustness of the proposed controller. The parameters of the crane under four working conditions are selected as the control objects, and the parameters of each working condition are shown in Table 9 below. The proposed VUFMS-PID controller, FUZZY-PID controller, and traditional PID controller are used to control the crane under the interference of side winds and environmental disturbances. The overshoot, positioning time, maximum swing angle, and swinging time of the load during crane positioning are used as evaluation indicators, and the simulation results are shown in the following Figures.

Table 9. Parameters of the crane under four working conditions.

Working Condition	Rope Length (m)	Payload Weight (kg)	Trolley Weight (kg)
1	1	5	10
2	1.5	5	10
3	1.5	8	10
4	1	8	10

In order to better demonstrate the real-time adjustment effect of the proposed control strategy on PID controller gains, Figure 14 shows the real-time output results of PID gains for the fuzzy controller of the trolley positioning system under environmental noise interference and working condition 1. From Figure 14, it can be seen that in order to reduce the impact of environmental noise on the control system, the output of PID gains also exhibits a pulsating change, similar to the environmental noise.

From Figures 15–22, it is observed that under the influence of strong wind and environmental noise disturbances, the crane controlled by the proposed VUFMS-PID controller exhibits the smallest maximum swing angle of the load under crosswind disturbance compared to the other two controllers, and it stabilized the quickest in each working condition. Additionally, the VUFMS-PID controller produced the smallest overshoot when positioning the trolley.

Through the above analysis, it is evident that the proposed VUFMS-PID controller exhibits excellent control performance in multiple working conditions and outperforms the other two controllers in handling crosswind and environmental noise disturbances. The fuzzy control system's domain adjustment makes the controller highly adaptable and robust, leading to better control precision and stability.

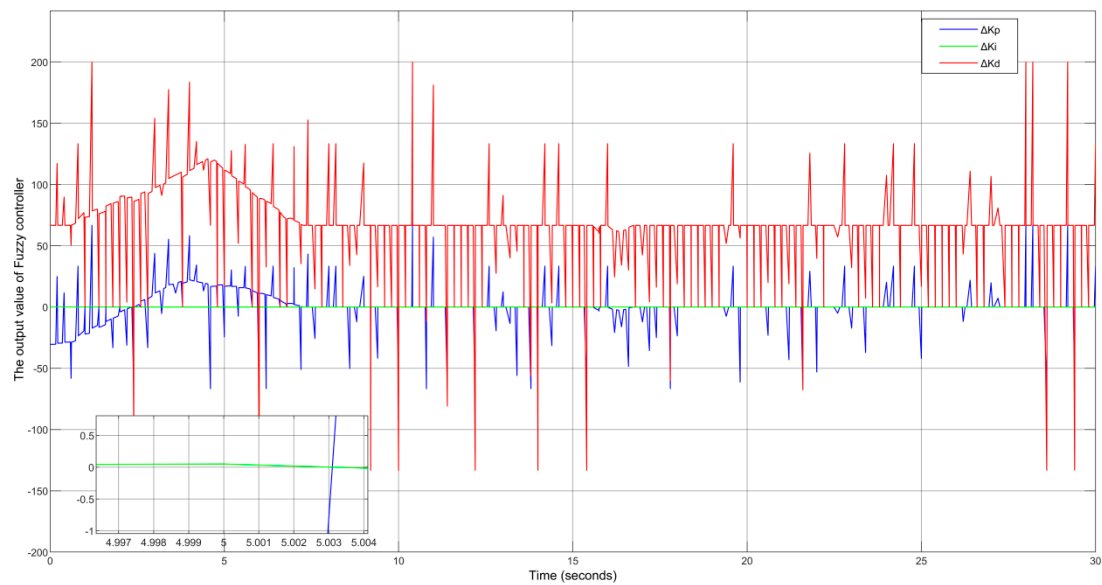


Figure 14. The output results of the PID gain of the fuzzy controller for trolley positioning under working condition 1.

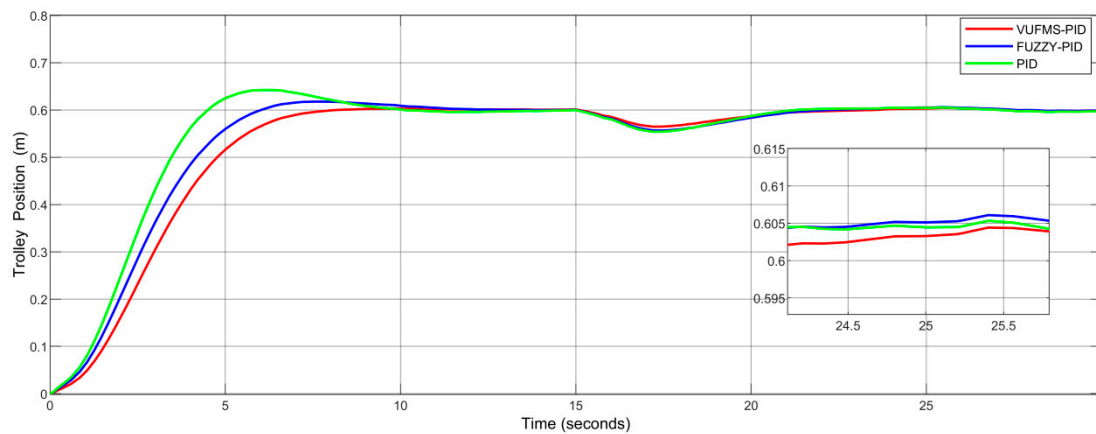


Figure 15. Trolley's position using VUFMS-PID, FUZZY-PID and PID controller under working condition 1 with a crosswind and white noise.

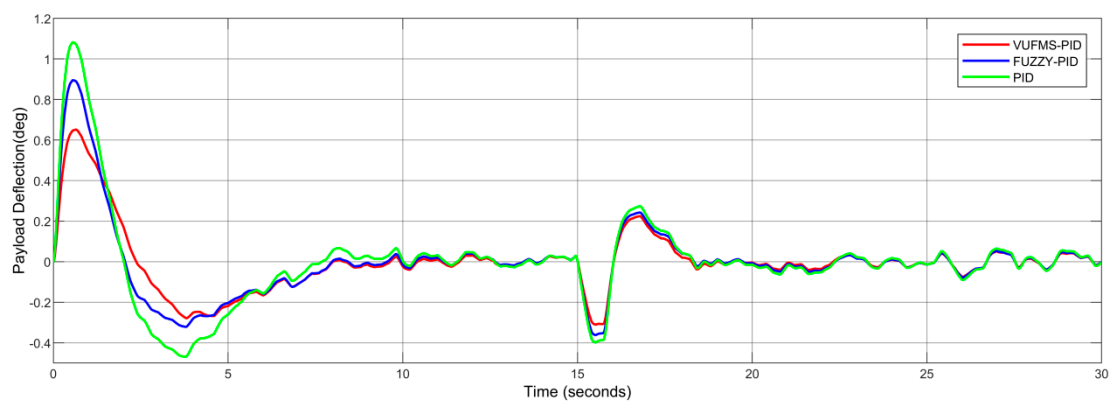


Figure 16. Payload's swing angle using VUFMS-PID, FUZZY-PID and PID controller under working condition 1 with a crosswind and white noise.

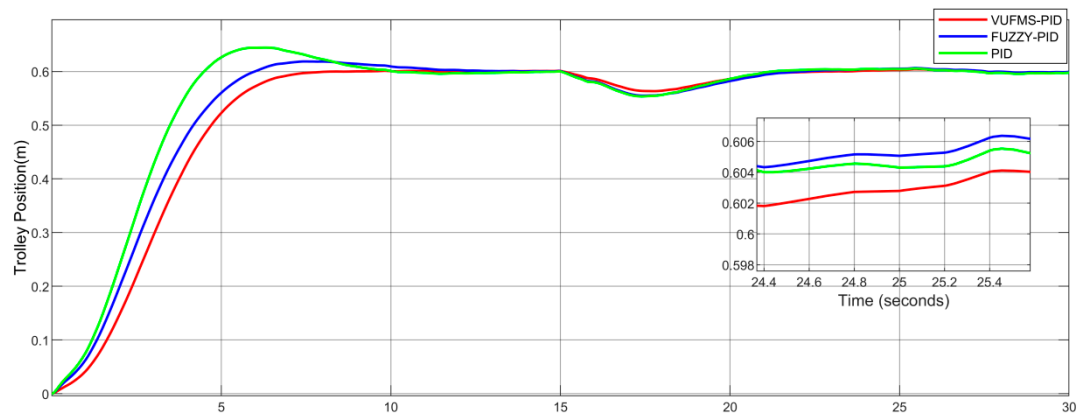


Figure 17. Trolley's position using VUFMS-PID, FUZZY-PID and PID controller under working condition 2 with a crosswind and white noise.

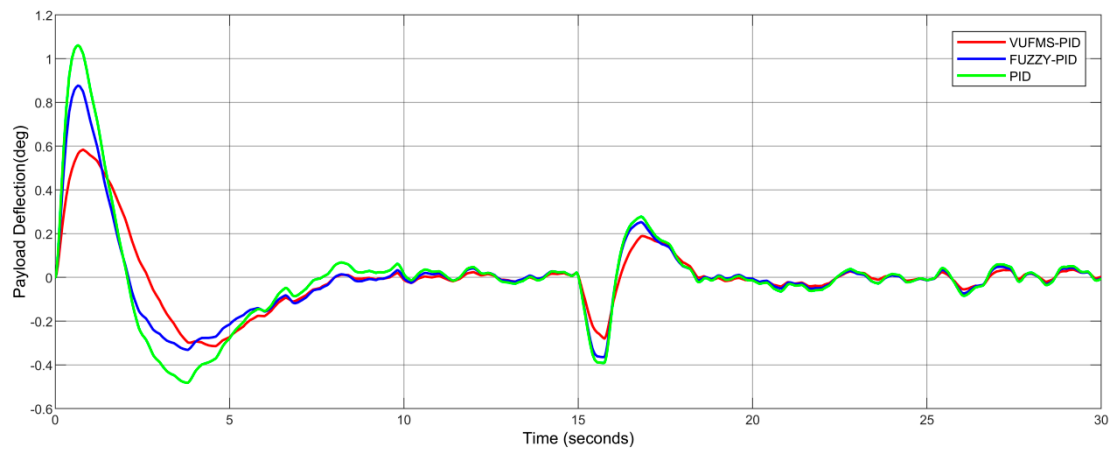


Figure 18. Payload's swing angle using VUFMS-PID, FUZZY-PID and PID controller under working condition 2 with a crosswind and white noise.

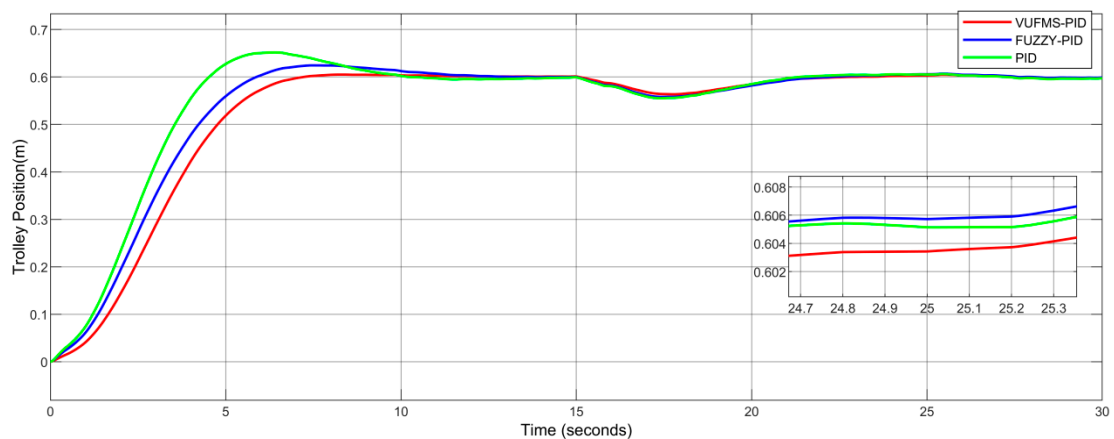


Figure 19. Trolley's position using VUFMS-PID, FUZZY-PID and PID controller under working condition 3 with a crosswind and white noise.

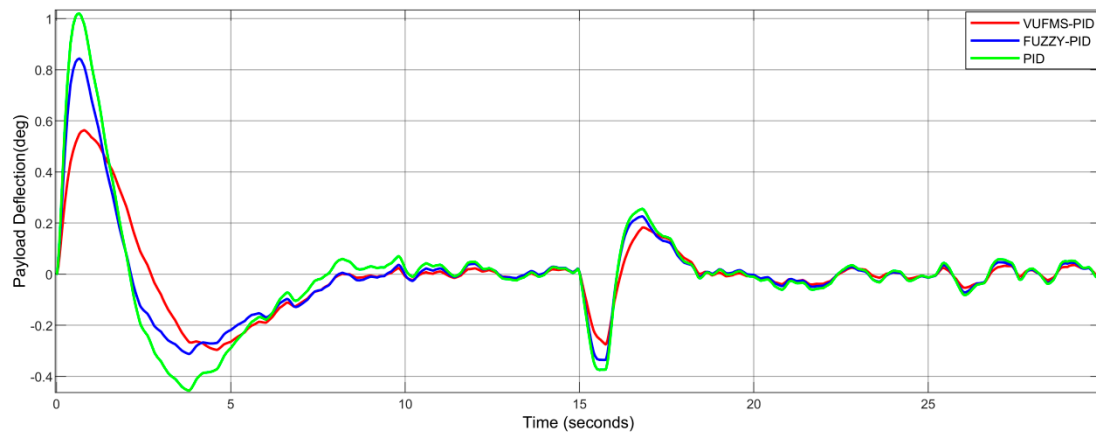


Figure 20. Payload's swing angle using VUFMS-PID, FUZZY-PID and PID controller under working condition 3 with a crosswind and white noise.

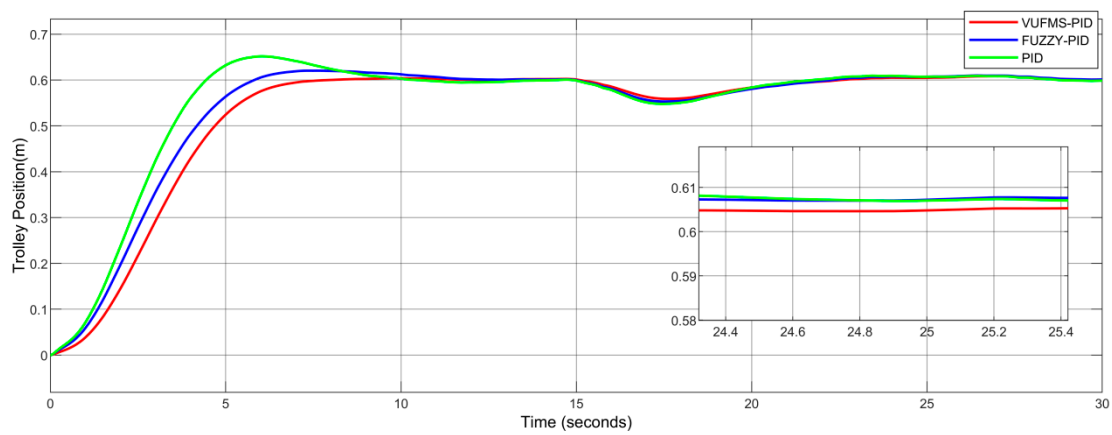


Figure 21. Trolley's position using VUFMS-PID, FUZZY-PID and PID controller under working condition 4 with a crosswind and white noise.

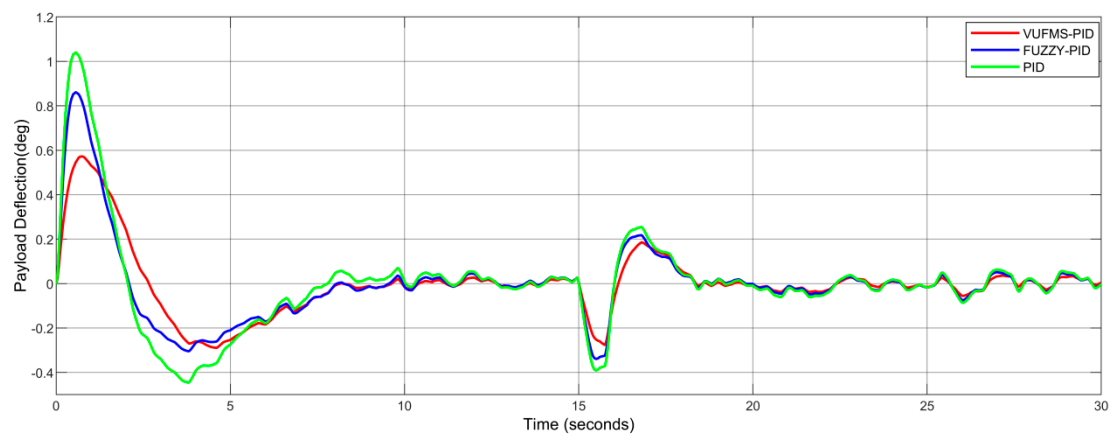


Figure 22. Payload's swing angle using VUFMS-PID, FUZZY-PID and PID controller under working condition 4 with a crosswind and white noise.

6. Conclusions

This paper presents a novel control strategy called the variable fuzzy domain PID control to enhance the trolley positioning and anti-sway control of bridge cranes. By monitoring the changes in the control system error and error variation rate, the control strategy generates appropriate scaling factors that dynamically adjust the domain of the

fuzzy controller. This approach effectively adds control rules to the fuzzy controller, thereby improving its accuracy and enhancing the stability and safety of bridge crane operations.

The results of the simulation demonstrate that the variable domain fuzzy PID control strategy surpasses the fuzzy PID and traditional PID control strategies regarding control accuracy, adaptability, and anti-interference capability. The trolley positioning control time is reduced by 29.1% and 28.2% compared to the other two methods, respectively. In anti-sway control, the maximum swing angle of the hanging weight is 0.66 deg which is reduced by 0.22 deg and 0.42 deg, respectively, compared to the other two methods. When the system is under four working conditions and increased environmental noise interference, the variable domain fuzzy PID control strategy used for anti-sway control of the hoisting load has a good result in the minimum swing angle and the shortest stabilization time. Moreover, when performing positioning control of the trolley, the strategy also leads to the smallest maximum overshoot. Therefore, this control strategy has good robustness and adaptability.

Author Contributions: Data curation, H.L. and J.M.; Investigation, H.L. and Q.W.; Methodology, Y.H. and H.W.; Built the Simulation program and was also responsible for writing the paper, H.L.; Writing—original draft, Y.H. and Y.Z.; Writing—review and editing; H.L. and Q.W. Supervision, Y.H., Y.Z. and H.L. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by Science and Technology Research Project of Henan Province, China (Grant No. 222102220080), Science and Technology Research Project of Henan Province, China (Joint Fund) (Grant No. 222103810083) and Science Foundation of Henan University of Technology (Grant No. 2019BS055).

Conflicts of Interest: The authors declare no conflict of interest.

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